



Merit, Quality & Excellence
Sukkur IBA University
Department of Computer Science

COURSE BINDER

CSC-301

Computer Networks

Fall 2025

PREPARED BY

CS Faculty

Course Syllabus – Computer Networks

Credit Units: 3+1	Semester : Fall-2025
Level: Year 2-3	Course code: CSC-301
Teaching Language: English	Status: Core
E-Mail: noornabi@iba-suk.edu.pk	Prerequisite: None
Office Hours: Will be specified according to Time table,	
Overall teaching period and hours: 16 weeks (2 lectures of 1.25 hours per week and 2.5 hours of labs per week)	

This introductory course in Data Communication and Networking provides foundational knowledge and practical understanding of how data is transmitted, processed, and managed across networks. It covers a wide range of topics including protocol architectures, basic networking concepts, network topologies, layered architectures and the OSI/TCP-IP models, physical layer functionality and transmission media, data link layer functionality, multiple access techniques, circuit and packet switching, LAN technologies, wireless networks, MAC addressing, networking devices, network layer protocols (IPv4 and IPv6), IP addressing and subnetting, routing protocols, transport layer protocols, ports and sockets, connection establishment, flow control, congestion control, application layer protocols, as well as the latest trends and emerging technologies in computer networks. The course equips students with both theoretical foundations and practical skills essential for understanding, designing, and working with modern computer networks.

Objectives

1. Describe the key terminologies and technologies of computer networks.
2. Explain the services and functions provided by each layer in the Internet protocol stack.
3. Identify various internetworking devices and protocols and their functions in a networking.
4. Analyze working and performance of key technologies, algorithms and protocols.
5. Build Computer Network on various Topologies

Learning outcomes

This course is mapped to contribute Program Outcome PO 1 and 2.

Learning outcomes achieved by students upon a successful completion of this course:

1. Analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies.
2. Specify and identify deficiencies in existing protocols, and then go onto formulate new and better protocols.
3. Analyze, specify and design the topological and routing strategies for an IP based networking infrastructure
4. Have a working knowledge of datagram and internet socket programming.

Evaluation Method

Mid-Term Exam 30% Lab Performance and Quiz Assignments 20%, Final Exam 50% (total 100%)

Assessments

Online quiz assignments which demand the students to comprehend the knowledge from textbook – MCQ (Multiple Choice Questions), True/False and short answers.

Mid and Final Exam where students must answer Short or Extended Response Questions from topics taught during semester. Student understanding of software engineering principles will be demonstrated by their performance on two tests and a final exam. Students are not allowed to use any learning materials during the exams.

For Labs and Lectures:

- Students must take notes during the lectures.
- It is mandatory that every student have a separate notebook for tutorial activities. Students without a notebook are assumed to be absent and lose marks reserved for tutorial activities.
- No electronic devices are allowed to be used during lectures and tutorials. In case, if a student is found using a mobile phone during a tutorial, he/she loses marks reserved for tutorial activities.
- Because class discussion is at the heart of this course, you are required to be in class, and what we do in the course of our class meetings will determine a large portion of your grade.
- Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials. During the lecture, I expect that all students are already familiar with previous lectures.

Quiz Assignments

- Every assignment will consist of True/False, MCQs, Short answers and Numerical questions
- Students are responsible for tracking their grades for all assignments, quizzes and exams.
- Students are expected to work independently during the quizzes and exams. Offering and accepting solutions from others is an act of plagiarism, which is a serious offense and all involved parties will be penalized. If any evidence of plagiarism or cheating is found, a student's grade for all quizzes will be nullified (canceled).
- Mandatory Assignments should be submitted by the indicated deadlines.
- Assignments which are submitted late will not be graded.

Lab Performance - Students will work on hardware, software and or a simulator aimed to apply and practice many of the topics covered in this course. Throughout the semester, students will produce a number of configurations, demonstrations, observations report etc.

Textbook and References

Core reading:

1. Kurose & Ross, "Computer Networking: A Top-Down Approach", 6th Edition Addison-Wesley, 2017. (Ross)

Additional:

2. Behrouz A. Forouzan William Stallings, "Data Communication and Networking", 5th Edition, McGraw Hill. (Behrouz)
3. Douglas Comer, "Computer Networks and Internets, 6th edition, Prentice Hall, 2006 (Comer)
4. William Stallings, "Data and Computer Communication", 10th edition, Pearson, 2013.
5. Peterson and Davie, "Computer Networks: A Systems Approach, 5th edition.
6. Andrew S. Tanenbaum, Computer Networks, Prentice Hall, 4th Edition, August 2002
7. Fred Hal Sall, "Data Communication, Computer Networks and Open Systems"
8. Larry L. Peterson and Bruce S, "Computer Networks A systems Approach"

Learning resources:

<https://skillsforall.com/>

Getting Started with Cisco Packet Tracer Network Addressing and Basic Troubleshooting Networking Devices and Initial Configuration Network Support and Security

[Jim Kurose Homepage \(umass.edu\)](http://jim.kurose.org/)

TENTATIVE COURSE OUTLINE/CLANDER

Week	Topic	Week Objectives / Reading / Lab
1	Introduction to Computer Networks, Overview of the course and syllabus. Philosophy and enablers of Computer Communication	To have been introduced with nature and need of Computer Networks To know and understand and use the Computer Networks Reading: Lecture Slides and Internet websites Lab-1 Networking Utilities
2	Chapter 01 Computer Networks and the Internet What is the Internet? The Network Edge The Network Core	To understand: A nuts-and-bolts and a services description. What is a protocol? Access networks, physical media, Forwarding, routing; packet switching; circuit switching; a network of networks Reading: Ross Chapter 1.1-1.3 Lab 2 - Networking in action – Wireshark Quiz Assignment 1
3	Performance: Delay, Loss and Throughput in Computer Networks Protocol layers and Their Service Models Networks under attack	To understand : Packet delay and loss, end-end throughput Layered architecture, encapsulation What can bad actors do? What defenses? Reading: Ross Chapter 1.4 -1.6 Lab 3 - Networking Tools and Equipment

4	Chapter 2 Application Layer Principles of Network Applications The Web and HTTP Email	To understand: Applications; client-server, P2P, sockets, APIs; transport services Overview, statelessness, HTTP messages, cookies, caching, HTTP/2 Email infrastructure (mail clients, SMTP servers), SMTP, IMAP Reading: Ross Chapter 2.1-2.3 Lab 4 - Networking Simulators – installation and usage Quiz Assignment 2
5	The Domain Name Service: DNS Peer-to-Peer File Distribution Video Streaming and Content Distribution Networks	To understand: The Domain Name Service: DNS Peer to-Peer File Distribution Video Streaming and Content Distribution Networks Reading: Ross Chapter 2.4-2.6 Lab 5- STP and RSTP
6	Socket Programming: Creating Network Applications	To understand: Socket abstraction, UDP and TCP socket programming Reading: Ross Chapter 2.7 Lab 6- Routing Configuration static Quiz Assignment 3

7	Chapter 3 Transport Layer Introduction and Transport-layer Services Multiplexing and Demultiplexing Connectionless Transport: UDP	To understand: Transport-layer services and protocols. Transport layer actions. What is multiplexing, demultiplexing? How is it done? How does it work in TCP and UDP? UDP segment structure. The Internet checksum. Reading: Ross Chapter 3.1-3.3 Routing Configuration static Lab 7- OSPF Single-Area Configuration
8	Principles of Reliable Data Transfer	To understand: Protocol mechanisms for reliable data transfer (rdt). Building an rdt protocol. Pipelining. Go-back-N. Selective Repeat. Reading: Ross Chapter 3.4 Lab 8- OSPF Multi-Area Configuration
9	Connection-oriented Transport: TCP Principles of Congestion Control TCP Congestion Control	To understand: The TCP connection and segment, RTT estimation and timeout, flow control Causes and costs of congestion, approaches to congestion control Classic TCP; Explicit Congestion Notification, delay-based TCP, fairness Reading: Ross Chapter 3.5-3.7 Lab 10- Configuring FTP Services Quiz Assignment 4

10	Chapter 4 Network Layer – Data Plane Network Layer Overview Whats Inside a Router?	To understand: Forwarding versus routing; data plane, control plane; network service model. Switching fabrics, buffering, and packet scheduling, net neutrality. Reading: Ross Chapter 4.1-4.2 Lab 11- Implementing DHCP
11	The Internet Protocol. Generalized Forwarding Middle boxes and Summary	To understand: IPv4, addressing, NAT, IPv6 Generalized forwarding, match plus action, Open Flow. Whats a middle box? Internet architecture, evaluation Reading: Ross Chapter 4.3-4.5 Lab 12- Ethernet Interface and Link Configuration Quiz Assignment 5
12	Chapter 5 Network Layer – Control Plane Introduction to the Network-layer control plane. Routing Algorithms. Intra-AS Routing in the Internet: OSPF.	To understand: Per-router versus SDN control plane. Link-State (LS) and distance-vector (DV) routing algorithms. The OSPF link-state routing protocol. Reading: Ross Chapter 5.1-5.3 Lab 13- VLAN Configuration

13	Routing Among the ISPs: BGP. The SDN Control Plane. ICMP: The Internet Control Message Protocol. Network Management and SNMP, NETCONF/YANG.	To understand: Role of BGP, advertising BGP paths, determining best paths, policy. SDN control plane, Open Flow, SDN controllers, data plane, control plane interaction. Network management framework, SNMP, MIBs, configuration management: NETCONF/YANG Reading: Ross Chapter 5.4-5.7 Lab 14- Inter-VLAN Routing Quiz Assignment 6
14	Chapter 6 Link Layer Introduction to the Link Layer. Error-Detection and -Correction Techniques. Multiple Access Links and Protocols	To understand: Link-layer: services, implementation context. Parity checks, checksums, cyclic redundancy check (CRC) Random Access, Taking Turns, and DOCSIS. Reading: Ross Chapter 61-6.3 Lab 15- ACL
15	Switched Local Area Networks. Data Center Networking. Retrospective: A Day in the Life of a Web Page Request.	To understand: Addressing, ARP, Ethernet, Link-layer switches, Virtual LANs. Data center architecture, trends A day in the life animation. Reading: Ross Chapter 6.4, 6.6, 6.7 Lab 16- NAT